- CLT Recap: $f_{X_1+X_2}(t) = f_{X_1} * f_{X_2}(t) = \int_{-\infty}^{\infty} f_{X_1}(t-z) f_{X_2}(z)\,dz$
  - for $X_1, X_2, \ldots$ iid $\rightarrow f_{X_1+X_2+\ldots X_n} = f_{X_1} * f_{X_2} \ldots * f_{X_n} = (f_{X_1})^{*n}$
  - looks like bell shaped curve with $E = nE[X_1]$ & $Var = nVar[X_1]$

- CLT says: Let $X_1, X_2 \ldots$ be iid with $E[X_1] = \mu < \infty$ & $Var[X_1] = \sigma^2 < \infty$
  - Then... $F_{(X_1+\ldots X_n - n\mu)/\sqrt{n\sigma^2}}(b) = F_{\mathcal{N}(0,1)}(b)$

$\mathcal{N}$ = normal

  - $\lim_{n\to\infty} P(\mathcal{N}(0,1) \le b) = \underbrace{1/\sqrt{2\pi} \int_{-\infty}^{b} e^{-x^2/2}\,dx}_{\text{CDF of bell-shaped curve}}$
  - $E = 0$, $Var = 1$, and the above integral $= 1$ (multiplied by $1/\sqrt{2\pi}$)

- Standard normal distro Z has $E = 0$, $Var = 1$
  - Let some distro $X = \sigma Z + \mu$, then $E[X] = \mu$, $Var[X] = \sigma^2$
  - $\mathcal{N}(\mu, \sigma^2)$ - Normal distro w/ $E = \mu$, $Var = \sigma^2$

    $\rightarrow 1/\sqrt{2\pi\sigma^2} \int_{-\infty}^{t} e^{-1/2((y-\mu)/\sigma)}\,dy$

- stability: if X_1 has $\mathcal{N}(\mu_1, \sigma_1^2)$ and X_2 has $\mathcal{N}(\mu_2, \sigma_2^2)$ and X_1, X_2 independent, then $X_1 + X_2$ has $\mathcal{N}(\mu_1+\mu_2, \sigma_1^2+\sigma_2^2)$
  - essentially: add together two stable distros, you'll get the same (another stable) distro

- I stockpiled 400 chocolate bars. Amt of time it takes to eat each is random w/ distro Exp(2 days). What is prob your stash lasts at least 6 months (182.5 days)?